

Stock Market Analysis

Comprehensive Investment Analysis Report

Project 5: Finance Domain Analysis

Report Date: September 08, 2026

Dataset: 252 Trading Days, 2023

Executive Summary

This report analyzes the supplied stock market dataset covering 252 trading days during 2023 to evaluate historical returns, volatility, drawdown, and risk-adjusted performance.

VERIFIED PERIOD STATISTICS:

- Period Return (Jan 2-Dec 19): 10.14%
- Starting Price (Jan 2): \$101.05
- Ending Price (Dec 19): \$111.30
- Highest Closing Price: \$117.74 (Nov 6)
- Lowest Closing Price: \$80.41 (May 26)
- Maximum Drawdown: -26.85%

RISK METRICS:

- Daily Volatility: 1.94%
- Annualized Volatility: 30.87%
- Sharpe Ratio: 0.47
- Up Days: 130 out of 251 (51.8%)
- Down Days: 121 out of 251 (48.2%)

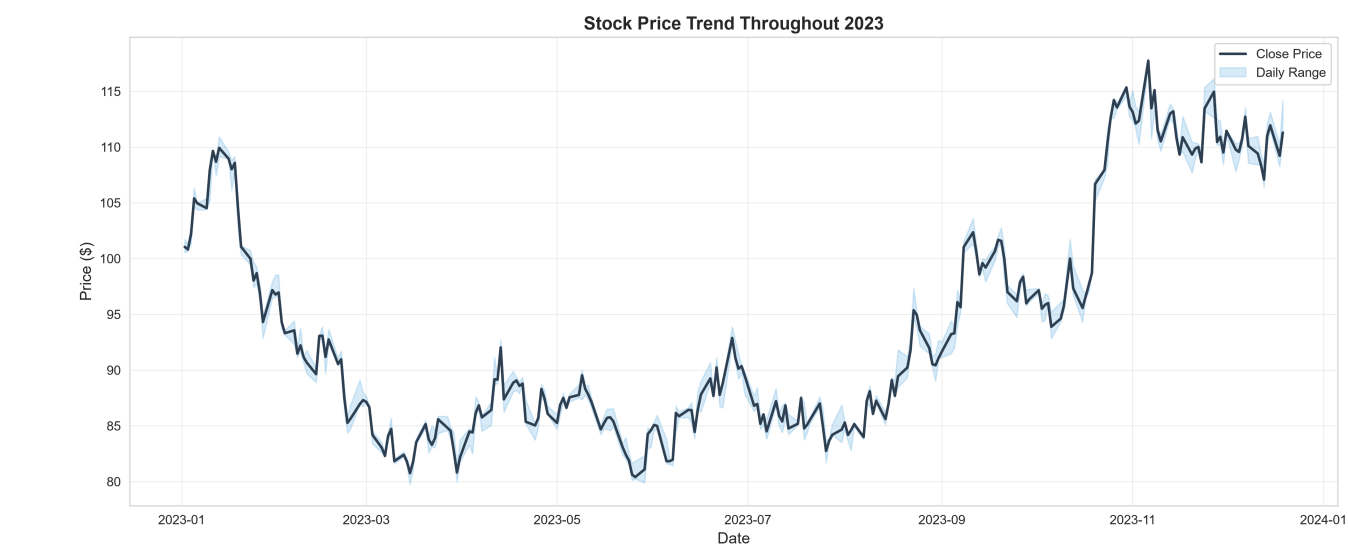
KEY FINDING:

The stock experienced a substantial drawdown of -26.85% in May followed by a strong recovery, ending the dataset period with a 10.14% positive return. October was the strongest month at +17.91%, while February was the weakest at -9.79%.

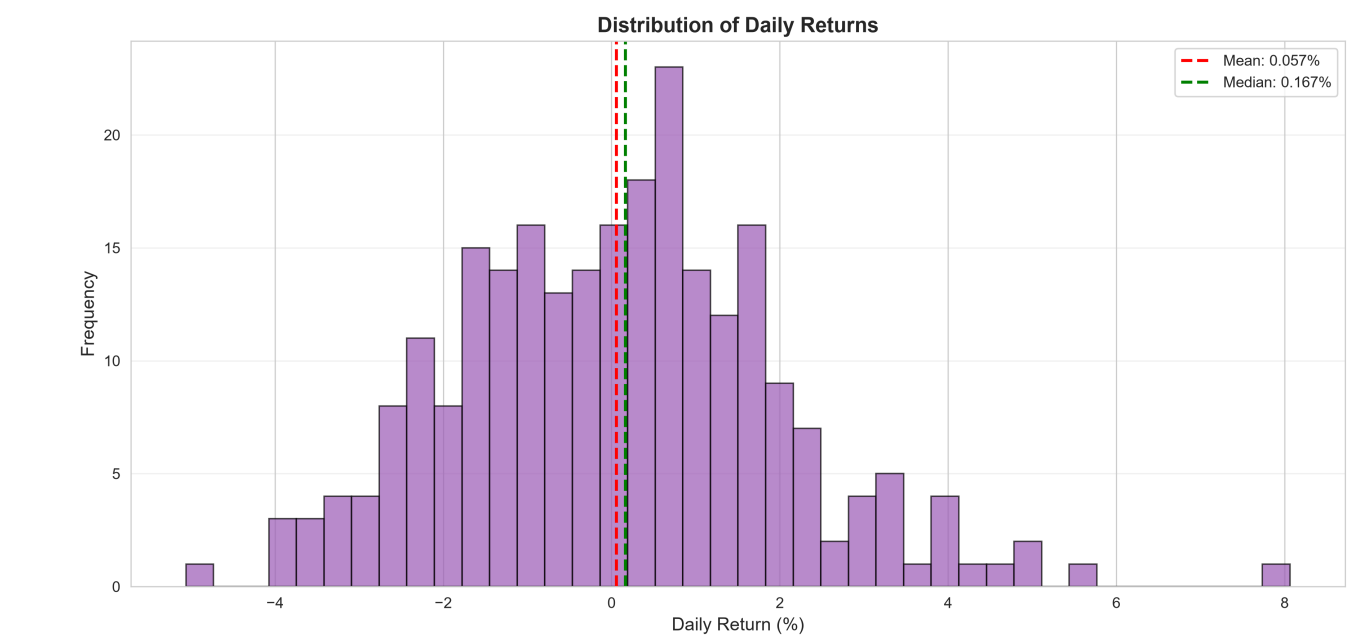
The analysis is based on the supplied dataset and should be interpreted as a historical examination rather than a prediction of future market performance.

Price and Returns Trends

Stock Price Trend During the Dataset Period

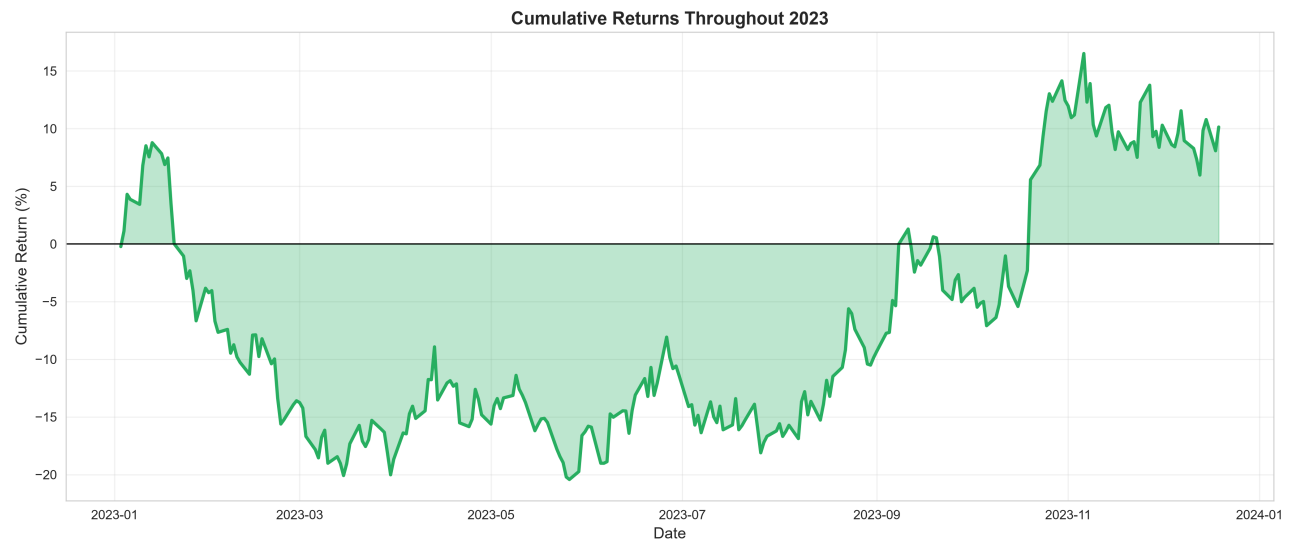


Distribution of Daily Returns

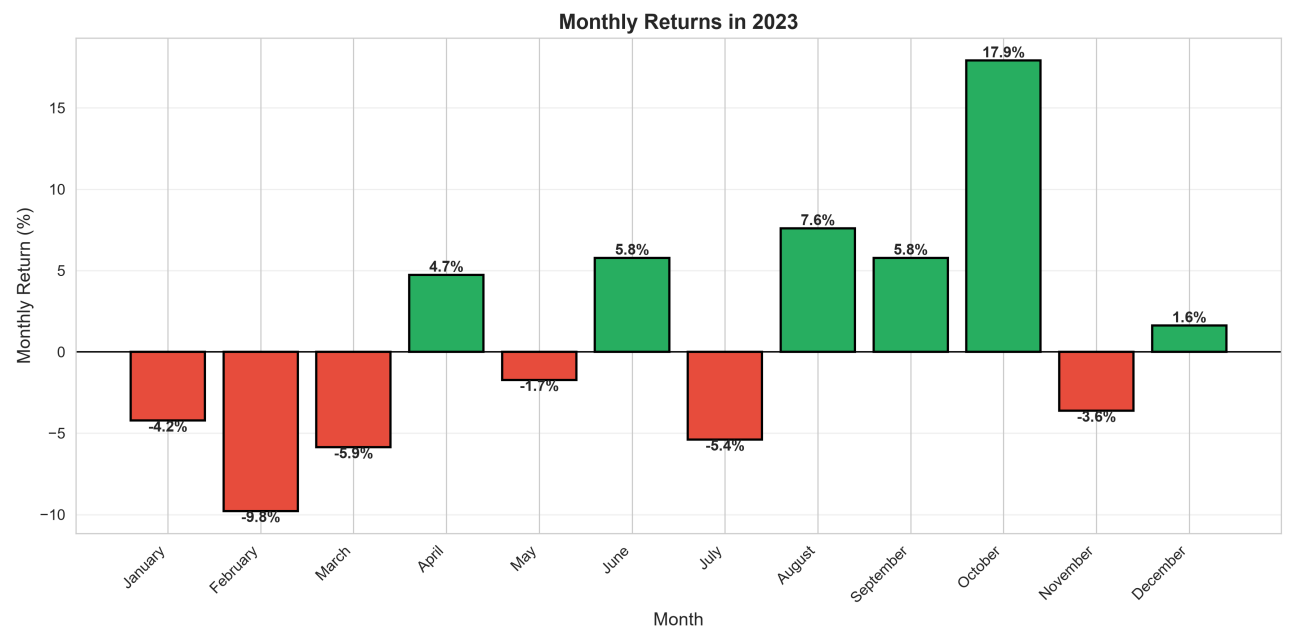


Cumulative Performance and Monthly Analysis

Cumulative Returns During the Dataset Period



Monthly Returns in the Dataset Period



Detailed Analysis

1. SUBSTANTIAL DRAWDOWN FOLLOWED BY RECOVERY

The stock declined to a lowest closing price of \$80.41 on May 26, corresponding to a maximum drawdown of -26.85% during the dataset period. The price subsequently recovered and reached the highest closing price of \$117.74 on November 6. The data therefore shows a substantial recovery after the May decline.

2. POSITIVE PERIOD RETURN DESPITE VOLATILITY

The stock increased from \$101.05 on January 2 to \$111.30 on December 19, resulting in a 10.14% period return. This positive ending return occurred despite the substantial mid-period drawdown.

3. OCTOBER WAS THE STRONGEST MONTH

October recorded the highest compounded monthly return at +17.91%, while February recorded the lowest at -9.79%. October therefore represented the strongest monthly performance in the analyzed dataset.

4. EARLY-YEAR WEAKNESS FOLLOWED BY STRONGER PERIODS

The first quarter recorded negative compounded monthly returns in January (-4.22%), February (-9.79%), and March (-5.87%), resulting in a compounded Q1 return of approximately -18.67%. Positive monthly returns were observed in April, June, August, September, October, and December, indicating stronger performance during several later periods.

5. POSITIVE RETURNS WITH MEANINGFUL DAILY VOLATILITY

Daily volatility was 1.94%, with annualized volatility of 30.87%. The median daily return of 0.167% was higher than the mean daily return of 0.057%, indicating that the distribution of daily returns was influenced by larger individual movements. Among the 251 days with valid daily returns, 130 were positive and 121 were negative.

6. MODEST RISK-ADJUSTED PERFORMANCE

The annualized Sharpe ratio was 0.47, calculated using a 0% risk-free rate assumption. This indicates positive but relatively modest risk-adjusted performance over the analyzed period. The result should be interpreted alongside the annualized volatility of 30.87%.

Investment Insights and Recommendations

HISTORICAL PRICE PATTERN:

The May low of \$80.41 marked the lowest closing price in the dataset, while the November high of \$117.74 marked the highest. The \$80-\$85 range can therefore be treated as a historical reference range rather than a confirmed future support level.

VOLATILITY CONSIDERATIONS:

Annualized volatility was 30.87%, indicating meaningful price variability during the analyzed period. The Sharpe ratio of 0.47 also indicates that the positive return was accompanied by substantial volatility.

QUARTERLY PERFORMANCE:

Q1 recorded a compounded return of approximately -18.67%, while Q2, Q3, and Q4 recorded approximately +8.89%, +7.68%, and +15.49%, respectively. These figures describe the observed dataset period and should not be assumed to represent a recurring seasonal pattern.

MONTHLY PERFORMANCE:

October was the strongest month at +17.91%, while February was the weakest at -9.79%. Other positive months included April, June, August, September, and December. These monthly movements describe historical performance rather than a reliable prediction of future returns.

FORWARD-LOOKING CONSIDERATIONS:

1. Historical Low: \$80.41 was the lowest closing price in the dataset.
2. Historical High: \$117.74 was the highest closing price in the dataset.
3. Volatility: Monitor whether future volatility remains close to the 2023 annualized level of 30.87%.
4. Risk Management: Future investment decisions should consider individual risk tolerance and objectives.

ANALYTICAL RECOMMENDATIONS:

- Monitor the historical \$80-\$85 price range as a reference area rather than assuming it will act as future support.
- Monitor the \$117-\$118 range as a historical high reference rather than a guaranteed resistance level.
- Compare future quarterly and monthly performance with the 2023 baseline before drawing conclusions about recurring patterns.
- Consider the combination of return, drawdown, volatility, and Sharpe ratio rather than evaluating performance using return alone.

Data Quality & Limitations

1. The dataset contains 252 trading-day records covering the supplied 2023 period, ending on December 19.
2. The supplied OHLC data contains inconsistencies in 78 rows:
 - 40 rows where High was below Open.
 - 38 rows where Low was above Open.

These conditions are inconsistent with standard OHLC relationships and may affect analyses that depend on the Open, High, and Low fields.

3. The original OHLC values were retained and were not automatically corrected, because the appropriate correction cannot be determined from the supplied dataset alone.
4. Daily Return and Cumulative Return calculations were validated against their corresponding price-based formulas, with no material mismatches above the validation threshold.
5. The Sharpe ratio uses a 0% risk-free rate assumption.
6. The analysis covers a single supplied annual period and therefore cannot establish long-term market trends or recurring seasonal behavior.
7. Historical returns, price levels, volatility, and drawdowns do not guarantee future performance.

Future analysis could use a longer time series and independently validated OHLC data to improve the reliability of technical and risk analysis.

Conclusion

The analyzed stock data shows a 10.14% positive period return, accompanied by substantial price volatility and a maximum drawdown of -26.85%. The stock declined to its lowest closing price of \$80.41 on May 26 before recovering to its highest closing price of \$117.74 on November 6.

Key findings include negative compounded returns during Q1, stronger performance during several later months, and a particularly strong October return of +17.91%. The annualized volatility of 30.87% and Sharpe ratio of 0.47 provide additional context for evaluating the return achieved during the dataset period.

The analysis should be interpreted as a historical examination of the supplied data. Observed monthly and quarterly patterns should not automatically be treated as recurring seasonal effects, and historical price levels should not be assumed to provide guaranteed future support or resistance.

The dataset also contains OHLC inconsistencies in 78 rows. These original values were retained, and this limitation should be considered when interpreting analyses involving Open, High, and Low prices.

Overall, the analysis provides a quantitative view of the stock's historical return, volatility, drawdown, and risk-adjusted performance while clearly documenting the available data-quality limitations.